BioLoop

WASTE-TO-VALUE DECISION LAB

Software Requirements Specification

Version 1.0 • Hackathon MVP Specification • 13 October 2026

BioValor’26 — WASTE TO WEALTH

Turning Biological Waste into Sustainable Wealth

Prepared as a development, testing, demonstration, and review reference.
This document describes a screening simulator, not a validated engineering design.

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Document Control

Project: BioLoop — Waste-to-Value Decision Lab

Document: Software Requirements Specification (SRS)

Version: 1.0

Status: Hackathon MVP specification

Prepared for: BioValor'26 — Waste to Wealth

Platform: Browser-based responsive web application

Implementation baseline: HTML, CSS, and client-side JavaScript

Date: 13 October 2026

Document purpose: Define the current MVP, expected behaviour, constraints, verification approach, and future evolution of BioLoop.

Field	Value
Product owner	Hackathon team (to be assigned)
Technical owner	Hackathon development team (to be assigned)
Primary users	Students, waste generators, farmers, early-stage project evaluators
Release target	One-day hackathon demonstrator
Approval	Team review required before claiming production readiness

Revision History

Version	Date	Change
0.1	13 Oct 2026	Initial prototype concept and baseline implementation
1.0	13 Oct 2026	Comprehensive SRS covering MVP behaviour, scientific model, quality attributes, edge cases, tests, and roadmap

1. Introduction

BioLoop is a browser-based decision-support simulator intended to help users explore the potential of organic waste for anaerobic digestion. It estimates methane volume, total biogas volume, gross methane energy, potential net electricity, and indicative daily economics from editable input assumptions.

The product is an educational screening tool and hackathon prototype. It is not a digester-sizing package, a laboratory measurement system, a safety controller, a regulatory compliance tool, or a guarantee of production or profit.

This SRS distinguishes implemented MVP requirements from recommended future requirements. A requirement marked FUTURE is not represented as already implemented.

1.1 Purpose

- Specify observable product behaviour so the team can build, test, demonstrate, and extend the prototype consistently.
- Define the inputs, calculations, outputs, constraints, and warnings used by the MVP.
- Give judges and reviewers a clear explanation of the simulator's scope and scientific limitations.
- Provide a traceable basis for acceptance tests and future development.

1.2 Scope

- Select a feedstock preset or use custom assumptions.
- Enter wet waste quantity, COD loading, COD removal, methane recovery, methane fraction, electrical efficiency, parasitic energy fraction, electricity value, and daily operating cost.
- Recalculate estimates in the browser as values change.

- Show methane, biogas, gross energy, net electricity, estimated electricity value, indicative daily net, and a qualitative cost-coverage message.
- Compare low, central, and high methane-recovery sensitivity cases.
- Explain the simplified process pathway and show limitations/disclaimers.
- Support responsive use on desktop, tablet, and mobile screens.

1.3 Out of Scope for MVP

- Live sensor ingestion, IoT connectivity, automatic temperature/pH control, and remote digester control.
- User accounts, authentication, cloud database, saved scenarios, collaboration, and audit trails.
- A validated kinetic digester model, hydraulic retention-time sizing, organic loading rate design, inhibition prediction, or full mass/energy balance.
- Automated optimisation of co-digestion mixtures or a claim that a given mixture is biologically optimal.
- Capital expenditure, financing, depreciation, taxes, subsidy eligibility, carbon-credit valuation, and full life-cycle assessment.
- Guaranteed digestate fertiliser quality, pathogen safety, or regulatory approval.
- Commercial engineering advice or plant safety instructions.

1.4 Definitions and Acronyms

Term	Definition
AD	Anaerobic digestion: microbial degradation of biodegradable material in the absence of oxygen.
Biogas	Gas produced during digestion, typically containing methane, carbon dioxide, and trace constituents.
COD	Chemical oxygen demand; a measure of oxygen-equivalent demand associated with oxidisable material.
COD loading	In this MVP, an assumed kg COD per kg of wet feedstock input. It is an editable screening parameter, not a universal feedstock property.
COD removed	Input COD multiplied by the assumed COD-removal fraction.
Methane recovery factor	A user-defined derating factor applied to the theoretical COD-to-methane relationship.
Methane fraction	Assumed volume fraction of methane in the total biogas.
Gross energy	Methane volume multiplied by the selected approximate lower heating value.
Net electricity	Gross methane energy multiplied by electrical conversion efficiency and by one minus the parasitic energy fraction.
Parasitic energy	Share of gross generated electricity assumed to be consumed by plant operations.
MVP	Minimum viable product.
Sensitivity case	An alternative model input used to show how outputs vary; it is not a statistical confidence interval.
SRS	Software Requirements Specification.

1.5 Intended Audience

- Hackathon developers implementing the interface and calculation engine.
- Bioprocess/biotechnology reviewers checking model assumptions and terminology.
- Judges evaluating application value, feasibility, and limitations.
- Future contributors who may add validation data, scenario saving, optimisation, or sensor support.

2. Overall Description

2.1 Product Perspective

BioLoop is a standalone client-side web application. The browser loads index.html, style.css, and script.js. The JavaScript reads the current form values, applies bounds, performs calculations, and updates visible result elements. The current design does not require a server, database, login, or external API.

The application is suitable for a local demo and static hosting. Hosting the files publicly does not turn the estimates into validated engineering results; data quality and model validation remain separate responsibilities.

2.2 Product Functions

- Feedstock preset selection: mixed canteen food waste, cattle manure/dung, crop residues (pretreated), and custom feedstock.
- Input handling for waste quantity, COD loading, COD removal, methane recovery, methane share, electrical efficiency, parasitic energy use, electricity value, and daily operating cost.
- Live calculation and presentation of estimated process outputs.
- Scenario comparison at 60% recovery, the selected central recovery, and 95% recovery.
- Reset assumptions to the original baseline.
- Display model-boundary notes, uncertainty warning, process pathway, and prototype disclaimer.

2.3 User Classes

User class	Goals	Knowledge assumptions
Hackathon demonstrator	Show inputs, calculations, and scenario response quickly	Basic browser use; may need a guided script
Student/research learner	Explore relationships between assumptions and outputs	Basic science/math; values may be illustrative
Farm/waste operator (exploratory)	Screen whether waste valorisation merits further study	May know daily waste quantities but may not know COD
Technical reviewer	Inspect model equations, boundaries, and assumptions	Can assess process science and model limitations
Judge/evaluator	Assess usefulness, feasibility, and responsible claims	Needs clear problem-to-value narrative

2.4 Operating Environment

- Modern evergreen browser: Chrome, Edge, Firefox, or Safari with JavaScript enabled.
- Desktop/laptop for the main demo; tablet and mobile browser for responsive use.
- Local files or static web hosting over HTTPS.
- No backend is required for current calculations.
- Internet is optional for core calculations; fonts or other externally hosted presentation assets, if any, may fall back when offline.

2.5 Design and Implementation Constraints

- Use HTML for structure, CSS for presentation, and plain JavaScript for the current MVP.
- No build tool or package manager is required for the current static version.
- The user must keep the three source files in the same folder unless paths are deliberately changed.
- All model values are entered or preset locally; no real-time laboratory or sensor data is available.
- The interface must make it clear that estimates depend on assumptions and are not measured results.

2.6 Assumptions and Dependencies

- The COD loading is interpreted as kg COD per kg wet feedstock; it must be measured or supported by a cited source before use for a real site.
- The theoretical methane factor of 0.35 m³ CH■/kg COD removed is an approximate stoichiometric relationship under standard conditions; actual yield depends on operating conditions and measurement basis.
- Methane recovery is used as an explicit derating factor in the MVP. It is not a measured plant recovery value.
- Methane share is treated as a volume fraction of total biogas.
- Methane lower heating value is approximated as 9.97 kWh/m³; actual energy varies with reference conditions and gas composition.
- Electricity value and operating cost are user-supplied screening assumptions.
- The MVP's daily net value excludes capital cost and numerous real-world costs and benefits.

3. System Features and Functional Requirements

3.1 Feedstock Selection

ID	Requirement	Status
FR-001	The system shall provide preset choices for mixed canteen food waste, cattle manure/dung, crop residues (pretreated), and custom feedstock.	MVP
FR-002	When a supported preset is selected, the system shall populate the corresponding illustrative COD loading, COD removal, methane recovery, and methane-share assumptions.	MVP
FR-003	The system shall allow the user to edit the preset values after selection.	MVP
FR-004	When COD loading is manually edited, the selected feedstock shall switch to Custom to avoid implying that the preset still applies.	MVP
FR-005	The interface shall identify preset assumptions as illustrative and request measured or source-supported values for real design work.	MVP

3.2 Input Validation and Bounds

ID	Requirement	Status
FR-006	The system shall accept wet waste input in kg/day and prevent negative values from producing negative process output.	MVP
FR-007	The COD loading input shall be bounded to 0–2 kg COD/kg wet feedstock by the calculation layer.	MVP
FR-008	The COD removal fraction shall be bounded to 0–0.95 in the calculation layer.	MVP
FR-009	The methane recovery factor shall be bounded to 0–1 in the calculation layer.	MVP
FR-010	The methane fraction shall be bounded to 0.01–1 in the calculation layer to prevent division by zero.	MVP
FR-011	Electrical efficiency shall be bounded to 0–0.60 and parasitic energy fraction to 0–0.80 in the calculation layer.	MVP
FR-012	Electricity value and operating cost shall not be negative in the model.	MVP
FR-013	The UI should visibly flag invalid or out-of-range user values instead of silently making every correction invisible.	Recommended improvement
FR-014	The UI should reject or clearly handle blank, non-numeric, Infinity, and extremely large values without breaking the page.	Recommended improvement

3.3 Calculation Engine

ID	Requirement	Status
FR-015	The system shall calculate input COD as wet waste mass multiplied by COD loading.	MVP
FR-016	The system shall calculate COD removed as input COD multiplied by the COD-removal fraction.	MVP
FR-017	The system shall calculate estimated methane as COD removed × 0.35 m^3/kg COD × methane recovery factor.	MVP
FR-018	The system shall calculate total biogas as estimated methane divided by methane fraction.	MVP
FR-019	The system shall calculate gross methane energy as methane volume × 9.97 kWh/m^3.	MVP
FR-020	The system shall calculate gross electricity as gross methane energy × electrical efficiency.	MVP
FR-021	The system shall calculate net electricity as gross electricity × (1 – parasitic energy fraction).	MVP
FR-022	The system shall calculate indicative electricity value as net electricity × electricity value per kWh.	MVP
FR-023	The system shall calculate indicative daily net as electricity value minus daily operating cost.	MVP
FR-024	The system shall recalculate outputs when a relevant input changes.	MVP
FR-025	The calculation layer shall use finite-number checks and safe bounds to avoid NaN/Infinity being presented as ordinary output.	MVP

3.4 Results and Interpretation

ID	Requirement	Status
FR-026	The system shall display estimated methane volume in m³/day.	MVP
FR-027	The system shall display estimated total biogas volume in m³/day.	MVP
FR-028	The system shall display gross methane energy in kWh/day.	MVP
FR-029	The system shall display net electricity in kWh/day.	MVP
FR-030	The system shall display estimated electricity value and indicative daily net in Indian rupees.	MVP
FR-031	The system shall provide a qualitative indication of whether estimated electricity value covers the entered daily operating cost before capital costs.	MVP
FR-032	The system shall label results as simulated estimates rather than measured plant outputs.	MVP
FR-033	The system should show the effective assumptions and calculation breakdown alongside the headline outputs.	Recommended improvement
FR-034	The system should clarify that a positive daily net is not project profitability or payback because capital expenditure and other costs are excluded.	MVP disclaimer / strengthen UI

3.5 Sensitivity Comparison

ID	Requirement	Status
FR-035	The system shall display low, central, and high methane recovery cases using 60%, the current selected recovery, and 95% respectively.	MVP
FR-036	The system shall recalculate scenario methane outputs when COD removed changes.	MVP
FR-037	The system shall label these cases as sensitivity scenarios, not a confidence interval or forecast probability.	MVP
FR-038	The system should allow users to configure the low and high case values in a future release.	Future

3.6 Reset and Navigation

ID	Requirement	Status
FR-039	The reset control shall restore all baseline input values, including the baseline feedstock selection.	MVP
FR-040	The system shall preserve the single-page workflow and allow users to move between input, results, process explanation, and disclaimer sections.	MVP
FR-041	The system should offer export/download of scenario inputs and outputs in CSV or JSON in a future release.	Future

3.7 Scientific Communication and Safety

ID	Requirement	Status
FR-042	The system shall explain that COD-based methane estimation does not model digester kinetics, retention time, inhibition, leakage, or actual plant performance.	MVP
FR-043	The system shall not describe digestate as safe fertiliser without appropriate testing, treatment, and regulatory review.	MVP
FR-044	The system shall not recommend burning waste as the only alternative or imply that digestion is always economically or environmentally superior.	MVP
FR-045	The system should display model version, reference conditions, units, and source notes for each scientific parameter in a future release.	Recommended improvement

4. Scientific and Mathematical Model

4.1 Model Boundary

The MVP is a steady-state screening calculator with daily input/output labels. It does not simulate time-dependent digester behaviour. The daily label assumes that the supplied feedstock quantity and parameter values represent a daily operating basis.

The model converts assumed COD removal into a theoretical methane volume, derates that volume using a user-defined recovery factor, infers total biogas from methane fraction, then estimates gross and net electricity and indicative value. It does not model heat demand, gas cleaning losses separately, methane leakage, flare use, digestate sale, water balance, or plant downtime.

4.2 Equations

Symbol / output	Equation	Units / interpretation
M	max(0, user-entered wet waste mass)	kg/day
L_COD	clamp(user COD loading, 0, 2)	kg COD/kg wet waste
COD_in	M × L_COD	kg COD/day
f_rem	clamp(COD removal fraction, 0, 0.95)	fraction
COD_removed	COD_in × f_rem	kg COD/day
R_CH4	clamp(recovery factor, 0, 1)	fraction; user assumption
V_CH4	COD_removed × 0.35 × R_CH4	m³ CH■/day; screening estimate
x_CH4	clamp(methane share, 0.01, 1)	fraction of biogas volume
V_biogas	V_CH4 / x_CH4	m³ biogas/day
E_gross	V_CH4 × 9.97	kWh/day; methane LHV approximation
η_elec	clamp(electrical efficiency, 0, 0.60)	fraction
E_elec,gross	E_gross × η_elec	kWh/day
f_parasitic	clamp(parasitic fraction, 0, 0.80)	fraction of generated electricity
E_elec,net	E_elec,gross × (1 – f_parasitic)	kWh/day
V_elec	E_elec,net × electricity value per kWh	■/day; avoided value assumption
N_daily	V_elec – daily operating cost	■/day; excludes capital and other omitted costs

4.3 Baseline Example (Illustrative)

Using the current default values: 500 kg/day wet food waste, COD loading 0.15 kg COD/kg wet waste, 70% COD removal, 85% recovery factor, 60% methane share, 35% electrical efficiency, 10% parasitic use, electricity value ■8/kWh, and operating cost ■1,200/day.

COD input = 500 × 0.15 = 75 kg COD/day. COD removed = 75 × 0.70 = 52.5 kg COD/day. Methane = 52.5 × 0.35 × 0.85 = approximately 15.64 m³ CH■/day. Biogas = 15.64 / 0.60 = approximately 26.07 m³/day. Gross methane energy = 15.64 × 9.97 = approximately 155.9 kWh/day. Gross electricity = 54.6 kWh/day. Net electricity after 10% parasitic use = approximately 49.1 kWh/day. Electricity value = approximately ■393/day. Indicative daily net = approximately –■807/day.

This result is useful because it demonstrates that an attractive waste-to-energy story is not automatically economically viable under every set of assumptions. It is an illustrative calculation, not a measured outcome.

4.4 Model Limitations and Scientific Cautions

- COD loading per kg wet waste varies substantially by feedstock, moisture, preparation, and analytical method. Presets must not be treated as universal literature values.
- The 0.35 m³ CH■/kg COD relationship is theoretical and depends on standard conditions and the meaning of COD removed. The recovery factor is a simplified lumped correction, not a mechanistic model.
- The methane fraction influences total biogas volume but does not independently change the methane volume estimated from COD.
- The model treats electricity value as a simple per-kWh value; real projects may use gas directly for heat or cooking, upgrade gas, export electricity, or displace another fuel.
- The model does not include heating demand, mixing, pumping, gas cleaning, flaring, leakage, digestate processing, transport, labour, maintenance, financing, capex, downtime, or seasonality.
- The low/central/high recovery display is deterministic sensitivity analysis, not a probability distribution or statistical confidence interval.
- Digestate quality and suitability depend on feedstock contaminants, pathogen risk, treatment, and local rules. Do not promise fertiliser revenue without testing.

4.5 Recommended Model Improvements

- Add explicit COD input/removal and methane calculations to a collapsible audit panel.
- Add units and tooltips explaining wet basis, COD basis, methane volume reference conditions, and electricity assumptions.

- Separate gas capture/recovery, methane yield, and conversion efficiency into parameters only when the team can explain each factor and avoid double-counting.
- Add scenario export with a timestamp, model version, inputs, outputs, and caveats.
- Add a data provenance field for each preset: source, sample type, measurement basis, region, and date.
- Validate model outputs against published datasets or laboratory/pilot results before making performance claims.

5. User Interface and Interaction Requirements

5.1 Main Layout

- Header/brand area identifies BioLoop and BioValor'26 context.
- Hero area explains the problem and purpose.
- Simulation workspace contains input controls and a results column.
- Economics panel summarises indicative daily value and net.
- Scenario comparison shows low, central, and high recovery assumptions.
- Process pathway explains collection/preparation, digestion, gas capture, and digestate management.
- Disclaimer and footer reinforce screening-only status.

5.2 Responsive Behaviour

Viewport class	Expected behaviour
Wide desktop (≥1200 CSS px)	Two-column workspace: inputs beside results; metric cards arranged in a grid.
Tablet (768–1199 CSS px)	Panels may stack or narrow while retaining readable controls and result labels.
Mobile (<768 CSS px)	Single-column layout; inputs and metrics stack; no horizontal page scrolling; touch targets remain usable.
Small mobile (<380 CSS px)	Labels and numerical outputs wrap without clipping; long units remain visible; buttons remain accessible.

5.3 Accessibility

- Every input must have a programmatically associated label.
- Interactive elements must be keyboard reachable and show a visible focus state.
- Colour must not be the only signal for positive/negative results or scenario distinctions.
- Dynamic output changes should be announced appropriately to assistive technology without excessive announcements.
- Text and controls should maintain readable contrast; text zoom should not break the layout.
- Use semantic headings, buttons, form controls, and meaningful accessible names.

5.4 User Feedback

- Show units next to every numerical field and result.
- Explain invalid or clamped values where possible; do not let a blank or malformed value silently appear scientifically meaningful.
- Keep the SIMULATED/ESTIMATED indicator visible near outputs.
- Display a clear empty/zero-output state when the waste quantity or COD loading is zero.
- Use clear wording for negative daily net: electricity value does not cover the entered operating cost under current assumptions.

6. Non-Functional Requirements

6.1 Performance

- NFR-001: Core calculations shall complete synchronously in the browser and update results within 200 ms under normal modern-device conditions.
- NFR-002: The static application should load without a backend and without mandatory network requests for calculation functionality.

- NFR-003: Repeated slider movement must not freeze or make the page unresponsive.

6.2 Reliability and Robustness

- NFR-004: Zero, boundary, and very large inputs must not cause an uncaught JavaScript error.
- NFR-005: The interface shall not present NaN or Infinity as a scientific result.
- NFR-006: Reset shall restore the defined baseline consistently.
- NFR-007: The model shall behave deterministically: the same inputs and version yield the same outputs.

6.3 Usability

- NFR-008: A first-time user should be able to change one assumption and identify the affected outputs without instruction from the developer.
- NFR-009: The main scenario and key results should be visible without navigating to another page.
- NFR-010: Scientific caveats should be readable and near the results they qualify.

6.4 Compatibility and Responsiveness

- NFR-011: Support current stable versions of Chrome, Edge, Firefox, and Safari where feasible.
- NFR-012: Support mobile screen widths from approximately 320 CSS px and up without horizontal overflow.
- NFR-013: The layout shall adapt to portrait and landscape orientation.

6.5 Maintainability

- NFR-014: Keep scientific constants named and documented rather than embedding unexplained literals throughout the code.
- NFR-015: Keep preset assumptions in a single configuration structure.
- NFR-016: Calculation functions should be separable from DOM rendering in a future refactor to enable unit tests.
- NFR-017: Changes to equations or default parameters must be recorded in a model-version note.

6.6 Security and Privacy

- NFR-018: The current application shall not request personal data or transmit scenario inputs to a server.
- NFR-019: User-controlled numeric inputs shall be parsed as numbers and shall not be interpreted as executable code.
- NFR-020: Any future persistence or analytics feature must disclose what is collected and obtain appropriate consent where required.
- NFR-021: If hosted publicly, use HTTPS and avoid unnecessary third-party scripts.

6.7 Availability and Offline Use

- NFR-022: Core calculations shall work when the source files are available locally and JavaScript is enabled.
- NFR-023: The product should degrade gracefully if optional fonts or external decorative assets cannot load.
- NFR-024: The MVP has no uptime guarantee or backend service-level objective.

7. Use Cases and Alternate Flows

UC-01 — Explore a feedstock preset

Primary actor: Student/operator. Preconditions: page loaded.

- 1. User selects a feedstock preset.
- 2. System loads its illustrative COD loading, COD removal, recovery, and methane-share values.
- 3. System recalculates outputs.
- 4. System displays the new scenario and caveats.
- Alternate/exception flows:
- If Custom is selected, no preset values are applied.

- If a preset value is manually edited, the feedstock is marked Custom when COD loading changes.

UC-02 — Enter a custom scenario

Primary actor: Technical user. Preconditions: page loaded.

- 1. User selects Custom or edits a parameter.
- 2. User enters daily wet waste, COD loading, removal, recovery, gas methane fraction, efficiency, parasitic use, electricity value, and operating cost.
- 3. System validates/bounds numeric values.
- 4. System recalculates metrics and scenarios.
- 5. User reviews the outputs and assumptions.
- Alternate/exception flows:
- Invalid values should be flagged in a future validation enhancement; calculation bounds must still protect the model.
- Zero waste or zero COD loading results in zero methane and energy.

UC-03 — Compare recovery sensitivity

Primary actor: Judge/student. Preconditions: a scenario exists.

- 1. User reviews low, central, and high recovery rows.
- 2. System shows methane volume for each case.
- 3. User changes waste quantity, COD loading, or removal.
- 4. All scenario values update based on the new COD removed.
- Alternate/exception flows:
- These values are sensitivity cases, not confidence intervals.

UC-04 — Evaluate indicative operating economics

Primary actor: Waste generator. Preconditions: scenario exists.

- 1. User enters local electricity value and daily operating cost.
- 2. System estimates net electricity and electricity value.
- 3. System subtracts daily operating cost.
- 4. System displays a positive-before-capital-costs or not-covering-cost message.
- Alternate/exception flows:
- If daily net is positive, system must not claim the whole project is profitable.
- If electricity value is zero, estimated electricity value is zero.

UC-05 — Reset the simulator

Primary actor: Any user. Preconditions: at least one input changed.

- 1. User activates Reset assumptions.
- 2. System restores original baseline input values.
- 3. System recalculates all outputs.
- 4. Baseline values and result state are displayed.
- Alternate/exception flows:
- Reset must not leave scenario bars or labels stale.

UC-06 — Present the project

Primary actor: Hackathon team member. Preconditions: prototype is loaded.

- 1. Presenter describes waste-to-value problem.
- 2. Presenter changes a realistic input.

- 3. Audience sees outputs update.
- 4. Presenter compares recovery cases.
- 5. Presenter explains economic limitations and validation plan.
- Alternate/exception flows:
- Presenter must not describe simulated outputs as measured results.

UC-07 — Use on a mobile device

Primary actor: Mobile visitor. Preconditions: URL or local copy available.

- 1. User opens the app on a phone.
- 2. Layout adapts to a narrow viewport.
- 3. User scrolls to inputs and changes controls.
- 4. Results update and remain readable.
- Alternate/exception flows:
- At narrow widths, long numbers and labels must wrap instead of clipping; if overflow occurs, treat as a defect.

8. Data Dictionary

Field / constant	Type	Default	Allowed / bound	Meaning
feedstock	enum	food	food/manure/crop/custom	Selected feedstock preset
wasteKg	number	500	≥ 0; input UI suggests up to 1,000,000	Wet feedstock kg/day
codLoad	number	0.15	0–2 in model	Assumed kg COD per kg wet feedstock
codRemoval	percent	70	0–95% in model	Fraction of input COD removed
recovery	percent	85	0–100% in model	Lumped derating/recovery factor
methaneShare	percent	60	1–100% in model; UI 45–75%	Methane fraction of total biogas
efficiency	fraction	0.35	0–0.60	Electrical conversion efficiency
parasitic	fraction	0.10	0–0.80	Share of gross electricity used by plant
electricityRate	■/kWh	8	≥ 0	Indicative electricity value
operatingCost	■/day	1200	≥ 0	Entered daily operating cost
COD_METHANE_FACTOR	constant	0.35	fixed in MVP	m³ CH■/kg COD removed theoretical relationship
METHANE_LHV_KWH_M3	constant	9.97	fixed in MVP	Approximate methane lower heating value, kWh/m³

9. Edge Cases and Error Handling

Case	Expected behaviour	Priority
Waste mass = 0	All gas and energy outputs equal zero; economics equal negative operating cost if cost > 0.	P1
COD loading = 0	COD input, COD removed, methane, biogas, and energy equal zero.	P1
COD removal = 0	Methane and energy equal zero.	P1
Methane recovery = 0	Methane and biogas equal zero; no divide-by-zero occurs because methane fraction remains bounded.	P1
Methane share approaches lower bound	Biogas rises according to methane/share ratio; UI should communicate sensitivity and avoid misleading precision.	P1
Efficiency = 0	Net electricity and electricity value equal zero.	P1
Parasitic use = 1 or above	Model clamp is 0.8; visible validation is recommended so the user understands the applied value.	P1
Negative waste input	Model clamps to zero; recommended UI should flag the invalid input.	P1
Negative electricity rate or operating cost	Model prevents negative values; recommended UI should flag the invalid input.	P1
Extremely large input	Output should remain finite or show a safe error; browser must not crash.	P1
Blank/non-numeric input	Current numeric parsing may interpret invalid values as zero; a future UX improvement should explicitly show an error.	P1
Offline/no font network	Core calculator works with fallback font.	P2
Mobile viewport 320px	No horizontal overflow or clipped controls.	P1
JavaScript disabled	Calculator cannot update; page should communicate that scripting is required in a future enhancement.	P2

10. Verification and Acceptance Test Plan

10.1 Acceptance Criteria

- AC-01: Selecting each preset populates the expected illustrative parameters.
- AC-02: Changing a numeric input or slider updates all dependent outputs without a page reload.
- AC-03: Reset restores the original baseline values and outputs.
- AC-04: With zero waste, methane, biogas, gross energy, gross electricity, and net electricity are zero.
- AC-05: With the baseline assumptions, calculated methane is approximately 15.64 m³/day, allowing for display rounding.
- AC-06: With fixed COD removed, the low-recovery methane result is lower than the central result when central recovery > 60%; the high case is higher when central recovery < 95%.
- AC-07: The UI labels outputs as simulated estimates and states that sensitivity cases are not confidence intervals.
- AC-08: At 320 CSS px, all controls and outputs are readable with no horizontal page scroll.
- AC-09: The app works from local files in a modern browser with JavaScript enabled.
- AC-10: No output or demo script claims guaranteed revenue, guaranteed yield, or digestate safety.

10.2 Test Cases

Test ID	Test setup/action	Expected result
TC-001	Load page with default values	Baseline inputs and calculated outputs appear; simulated indicator visible.
TC-002	Use baseline inputs	COD in = 75 kg/day; COD removed = 52.5 kg/day; methane ≈ 15.64 m³/day.
TC-003	Set waste to 0	Gas and energy outputs become 0; no NaN/Infinity.
TC-004	Set COD loading to 0	Gas and energy outputs become 0.
TC-005	Set COD removal to minimum UI value	Outputs reduce consistently.
TC-006	Change recovery from 85% to 60%	Methane falls proportionally with recovery; scenario rows update.
TC-007	Change methane share from 60% to 50%	Methane unchanged; total biogas increases.
TC-008	Set electrical efficiency to 0	Electricity and electricity value become 0.
TC-009	Increase parasitic fraction	Net electricity decreases.
TC-010	Set electricity value to 0	Estimated electricity value is ■0; daily net equals negative operating cost.
TC-011	Set operating cost to 0	Indicative daily net equals electricity value.
TC-012	Select manure preset	Preset values are loaded and outputs recalculate.
TC-013	Edit COD loading after preset selection	Feedstock selector changes to Custom.
TC-014	Click reset after multiple edits	All baseline values return.
TC-015	Test viewport at 320px	No horizontal overflow; controls are usable.
TC-016	Test keyboard navigation	Inputs and reset button are keyboard reachable; focus visible.
TC-017	Disable internet after app loaded locally	Core calculations still work.
TC-018	Enter extreme values within/above UI range	No uncaught errors; outputs remain finite or error handling is clear.
TC-019	Check economic message with negative net	Message says operating cost is not covered; no profit claim.
TC-020	Review disclaimer	Scientific and economic limitations are visible and understandable.

10.3 Test Evidence

- Capture desktop screenshot with baseline values.
- Capture mobile screenshots at 320–375 px and tablet width.
- Record baseline calculation independently in a spreadsheet or test script.
- Record browser/version, test date, expected output, observed output, and pass/fail for each test.
- Any discrepancy in formulas, units, or labels should be fixed before the public demo.

11. Deployment and Operations

11.1 Local Deployment

- Keep index.html, style.css, and script.js in the same directory.
- Open index.html in a modern browser.

- Confirm JavaScript is enabled and no browser-console errors occur.
- Run the acceptance smoke tests before presenting.

11.2 Static Hosting

- Upload the three files to a static host or repository.
- Preserve file names and relative paths exactly.
- Use HTTPS for a public deployment.
- Test the deployed URL on desktop and a real phone; browser emulation alone is not a substitute for final mobile verification.
- Avoid claiming the prototype has been visually verified on all devices until those tests have been performed.

11.3 Backup and Versioning

- Keep a known-good ZIP archive before making changes.
- Use Git commits for each meaningful feature or formula change.
- Record changes to scientific constants and presets in a model-change log.
- Use version labels in the interface and the SRS to keep demo claims aligned with implementation.

12. Risks, Mitigations, and Ethical Considerations

Risk	Impact	Mitigation
Illustrative COD presets mistaken for measured data	Severely misleading output	Label presets; provide source/provenance; require measurement for real decisions.
Theoretical methane factor interpreted as guaranteed yield	Overstated production estimate	Explain theoretical basis and recovery derating; validate with tests.
Operating-cost result interpreted as profit	Poor investment decisions	Use 'indicative daily net'; explicitly exclude capital and omitted costs.
Digestate automatically treated as safe fertiliser	Environmental/health risk	Require testing, treatment, and local regulatory review.
Unmodelled methane leakage or process energy	Overstated climate/energy benefit	List omitted processes and add them to later lifecycle analysis.
User enters inconsistent units	Incorrect calculations	Show units inline; add unit validation and source notes.
Mobile layout clips controls	Demo failure and poor accessibility	Test narrow widths and real devices before presentation.
Time pressure during hackathon	Incomplete features or broken demo	Freeze MVP scope; preserve a known-good build; rehearse a deterministic scenario.
No physical validation	Limited scientific credibility	Describe prototype as a screening simulator and propose a lab/pilot validation plan.

13. Demo Plan for BioValor’26

13.1 Five-Minute Demonstration

Time	Action	Message
0:00–0:40	State problem and introduce BioLoop	Organic waste can be a resource, but potential and economics are often unclear before investment.
0:40–1:20	Select food waste and show input assumptions	The model makes assumptions visible instead of hiding them.
1:20–2:10	Change daily waste or COD removal	Outputs update instantly; the model connects organic loading to methane and energy.
2:10–3:00	Compare low/central/high recovery cases	A single number can create false confidence; sensitivity analysis exposes uncertainty.
3:00–4:00	Change electricity value and operating cost	Energy potential does not automatically mean economic viability.
4:00–5:00	Explain limitations and next steps	Next: validate feedstock data, add provenance, pilot measurements, and safe digestate assessment.

13.2 Questions Judges May Ask

Question	Recommended honest answer
Is the methane output experimentally validated?	No. The current output is a COD-based screening estimate. Validation against literature data and pilot measurements is a planned next step.
Why use 0.35 m³/kg COD?	It is an approximate theoretical methane relationship under standard conditions; the model applies a user-selected recovery factor to avoid presenting the theoretical value as guaranteed output.
Why might a positive result still fail economically?	The MVP excludes capital expenditure, financing, maintenance variability, downtime, heat demand, gas cleaning, digestate management, and other costs.
Can it optimise a mixture of waste types?	Not in the current release. It can compare user-defined assumptions, but a genuine mixture optimiser requires compatible feedstock data and constraints.
Does it measure pH or temperature?	No. There are no connected sensors in the current browser prototype.
Is the digestate safe to apply to soil?	The tool makes no such guarantee. Quality, pathogens, contaminants, treatment, and applicable requirements must be assessed.
Why not use Python?	The current calculations are small and deterministic, so JavaScript is sufficient in-browser. Python could be useful later for data analysis, calibration, optimisation, or model fitting.

14. Future Roadmap

Phase	Candidate feature	Value	Dependencies
P1 — MVP hardening	Visible validation, model audit panel, source notes, responsive QA	Improves trust and demo reliability	Current HTML/JS/CSS
P2 — Scenario handling	Save/export CSV or JSON, compare named scenarios	Makes results reusable	Defined schema and model versioning
P3 — Feedstock intelligence	Multi-feedstock mass/COD mixture model with provenance	Supports more realistic screening	Comparable feedstock data and constraints
P4 — Economics	Capex, payback, financing, digestate handling, gas use pathways	Improves investment screening	Locally sourced costs and validated equations
P5 — Scientific validation	Benchmark mode against lab/pilot datasets	Measures prediction error	Quality-controlled data
P6 — Advanced model	Temperature, retention time, loading rate, inhibition and kinetics	More realistic process analysis	Bioprocess expertise and validation
P7 — Sensor integration	pH/temperature/flow sensor dashboard	Live monitoring	Hardware, backend/security, calibration
P8 — Responsible impact analysis	Methane leakage, displaced energy, transport, digestate and lifecycle boundaries	Better environmental accounting	Defined LCA method and defensible data

15. Requirements Traceability Matrix

Objective	Related requirements	Verification
Estimate methane	FR-015 to FR-017	TC-002, TC-003, TC-004, TC-006
Estimate total biogas	FR-018, FR-027	TC-007 and methane-share boundary tests
Estimate energy	FR-019 to FR-021, FR-028 to FR-029	TC-008, TC-009
Estimate indicative economics	FR-022, FR-023, FR-030, FR-031, FR-034	TC-010, TC-011, TC-019
Compare uncertainty	FR-035 to FR-038	TC-006 and scenario comparison test
Support feedstock selection	FR-001 to FR-005	TC-012, TC-013
Reset assumptions	FR-039	TC-014
Mobile accessibility	NFR-011 to NFR-013 and UI section 5	TC-015 and real-device check
Scientific honesty	FR-032, FR-037, FR-042 to FR-045	TC-020 and judge-QA review
Offline core	NFR-022 to NFR-024	TC-017

16. Acceptance and Sign-Off

The MVP may be considered ready for hackathon demonstration when all P1 acceptance criteria pass, the default calculation is independently checked, the core interactions work, and mobile layout has been tested at narrow widths. Any untested behaviour must be described as unverified rather than complete.

A technical/bioprocess reviewer should review the equations, units, preset provenance, and limitations before the prototype is used to influence real investment or process decisions.

Role	Name	Decision	Date
Product/team representative	_____	Approve / revise	_____
Technical reviewer	_____	Approve / revise	_____
Bioprocess reviewer	_____	Approve / revise	_____

Appendix A — Implementation Baseline

- index.html: page structure, labels, feedstock selector, input controls, result panels, scenario list container, process pathway, and disclaimers.
- style.css: visual system and responsive layout rules.
- script.js: preset configuration, numeric bounds, COD/methane/energy/economic calculations, result rendering, sensitivity comparison, and reset behaviour.
- No Python runtime is required for the current browser-based implementation.
- No backend, database, login, sensor connection, or persistence layer is implemented in the MVP.

Appendix B — Known Gaps to Check Against the Running Build

- Confirm all mobile breakpoints and no horizontal overflow on an actual device.
- Check whether every invalid numeric entry is visibly explained; current calculation clamps values but may not provide complete field-level error messaging.
- Confirm output rounding and labels are consistent with units and the daily basis.
- Confirm the low/high recovery scenarios behave sensibly when the central recovery is below 60% or above 95%; current scenario ordering is by named case, not sorted value.
- Confirm external font requests are optional and do not prevent offline operation.

• Add data provenance before presenting feedstock preset values as evidence-based values.

Appendix C — Glossary for Presentation

Plain-language phrase	Meaning
Waste input	Mass of wet biological material entered per day.
COD loading	Assumed amount of chemically oxidisable material represented as COD per kg of wet feedstock.
COD removal	Share of the assumed input COD that is removed during treatment.
Methane recovery	A simplified factor that derates theoretical methane potential.
Biogas	The combined gas stream; methane is only one component.
Gross energy	Energy contained in estimated methane before conversion losses.
Net electricity	Estimated electricity after conversion efficiency and assumed plant electricity use.
Indicative daily net	Estimated electricity value minus the entered daily operating cost; not full project profit.
Sensitivity analysis	Showing how the result changes when an assumption changes.